

P7-T08

Literature-calibrated synthetic golden cases

Detailed case report with comparison plots, tables, provenance, and admission limits

Outcome

The package is a deterministic, project-authored synthetic authority for Golden-data audit and report regression. It is calibrated to published summary values and an independent unit identity; it does not claim that DRAGON/DONJON/SERPENT inputs, nuclear data, or source outputs were reproduced.

Artifact	Value
Status	Approved/ApprovedGolden
Coverage	SyntheticLiteratureCalibrated
Cases	4
Approved artifact SHA-256	b4d46cf2535c3616622c4c0a15ab1fcc6979662d3f1e02c25f4de0e1c61228ca
Definition SHA-256	d93a8ddf8f34e184584f8a7718921503294bffacfa153b440191bd6beaefeb17
Generated bytes	211,470

Reading rule

Values labelled **ReportedTarget** are source claims, not tolerances. Values labelled **ComputedTheory** are independently calculated identities. Values labelled **SyntheticCalibrated** are generated from the frozen definition. No row is a direct external solver reproduction.

1. Case inventory and authority boundary

The four cases intentionally cover lattice and full-core shapes while remaining outside the runtime physics authority boundary. The paper locators are preserved in the generated JSON and repeated here for auditability.

Case	Model shape	Primary comparison	Disposition
p7-t08-s4-lattice-37-bundle-kcross-v1	S4 37-element lattice DRAGON/SERPENT k-effective cross-check	SyntheticOnlyReportedTargetComparison	Synthetic only; no Core mapping
p7-t08-s5-lattice-depletion-300d-v1	S5 constant-power lattice depletion and burnup handoff	SyntheticOnlyReportedTargetComparison	Synthetic only; no Core mapping
p7-t08-s1-fullcore-atf-summary-v1	S1 CANDU-6 full-core schedule and alternate-cladding summary	SyntheticOnlyReportedTargetComparison	Synthetic only; no Core mapping
p7-t08-s4-fullcore-refueling-summary-v1	S4 full-core refueling aggregate comparison	SyntheticOnlyReportedTargetComparison	Synthetic only; no Core mapping

Source artifact boundary

The official records expose the papers, but the complete solver package is not present. In the accessible Holmes thesis, the refuelling script names external CPO, Data, Results, procedure, and archive directories plus `Candu6\_\*.x2m` and `.result` files. The S1 and S5 records likewise establish the study and headline quantities without a repository-compatible input/output manifest. The package therefore preserves the P7-T07 admission gap.

Licensing decision: facts-only attribution. The approved artifact contains project-authored synthetic values and concise attributed facts; source bytes, figures, full tables, input decks, nuclear data, and source-result payloads remain external. No derived source run is claimed. Permission is still required for digitization, full-table redistribution, or raw input/output bundles.

Source	Official record	License / use
S1: CANDU-6 operation simulations using accident tolerant cladding candidates	https://doi.org/10.1016/j.anucene.2018.10.026	CC BY-NC-ND 4.0; source artifact remains external Small reported summary quantities only; no derived source output or input deck is redistributed.
S4: Automated Refueling Simulations of a CANDU for the Exploitation of Thorium Fuels	https://publications.polymtl.ca/1307/	Open access repository copy; copyright retained by author Small reported table aggregates only; no source script, database, or full output table is redistributed.
S5: Neutronic analysis for accident tolerant cladding candidates in CANDU-6 reactors	https://doi.org/10.1016/j.anucene.2017.11.016	CC BY 4.0; source artifact remains external Reported cycle inputs and headline exit-burnup value only; exact external run is not admitted.

2. Case A - S4 lattice k-effective cross-check

The Holmes thesis reports a 37-element CANDU lattice comparison between DRAGON and SERPENT. The synthetic output is the minimax midpoint of the two printed k-effective values. This is a transparent calibration rule, not a claim that either external solver was run.

Source locators: Holmes thesis printed pp. 17-18, Table 3.1; Annex 2 printed pp. 47-55.

S4 lattice k-effective: printed values and synthetic midpoint

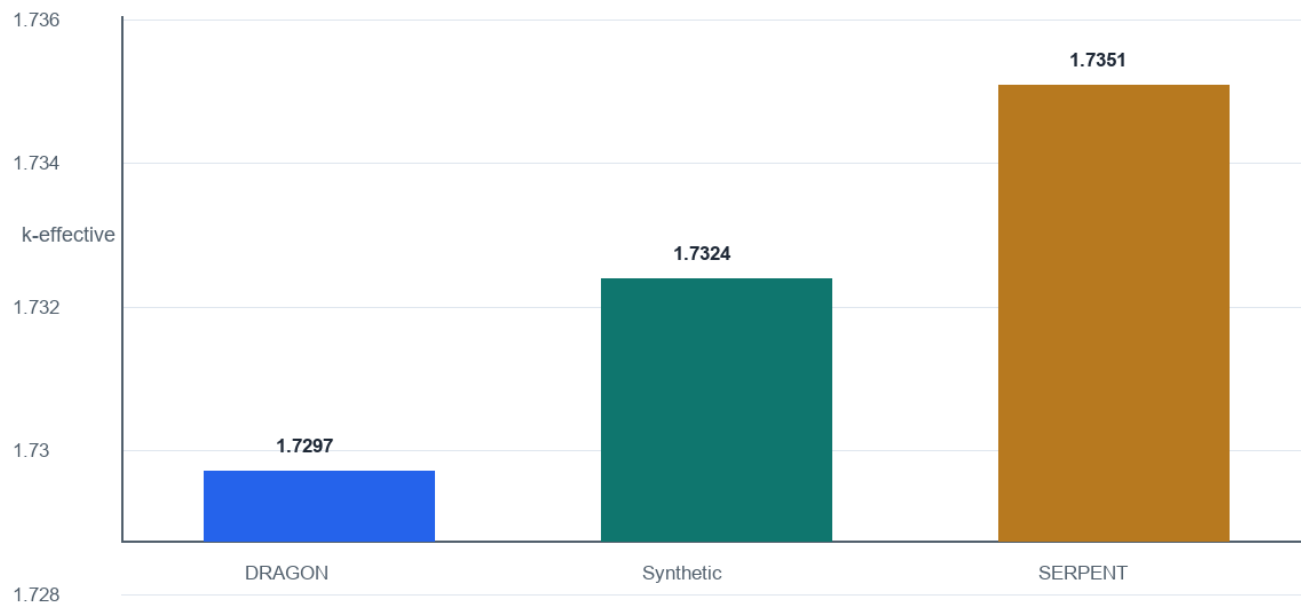


Figure 1. The synthetic midpoint minimizes the maximum absolute difference to the two printed method values.

Observable	Reported target	Synthetic	Signed diff	Relative diff
S4-R06/DRAGON4	1.72972	1.7324	2.685 mk	0.1552%
S4-R06/SERPENT	1.73509	1.7324	-2.685 mk	0.1547%

Interpretation: the printed pair spans 5.37 mk. The midpoint is 2.685 mk from each method. The source's printed error label is retained as literature evidence only; it is not promoted to a repository tolerance.

3. Case B - S5 lattice depletion and burnup handoff

The source reports a 300-day constant-power cycle at 31.9713 kW/kg and a rounded exit burnup of 9.6 GWd/t. The generator independently evaluates the unit identity and emits a complete day-by-day synthetic history.

S5 constant-power burnup identity

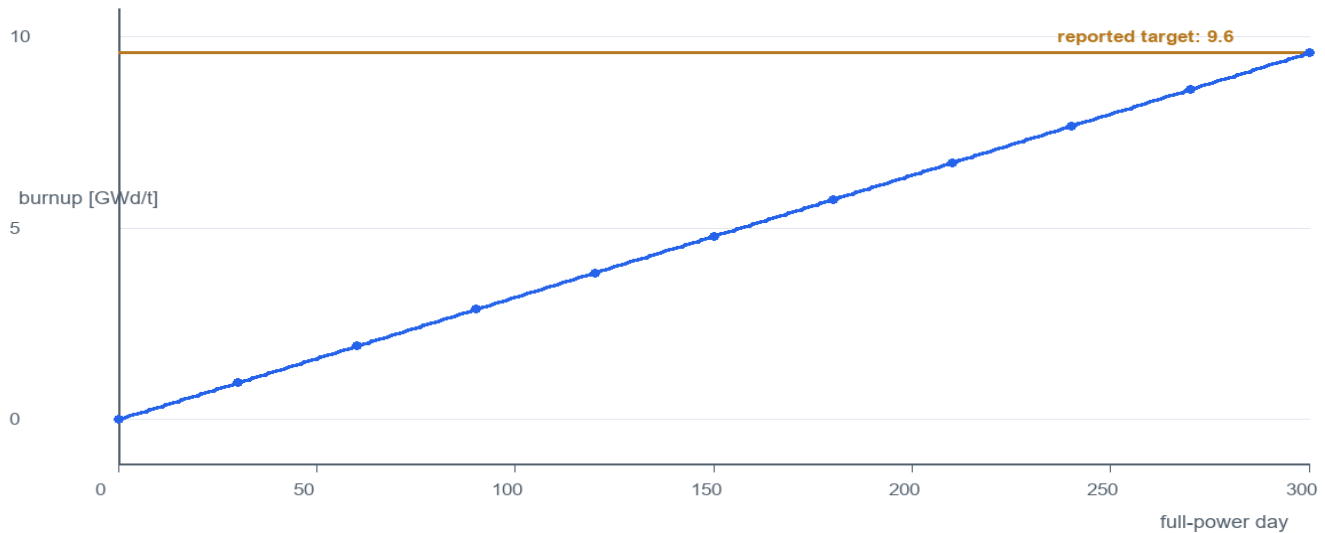


Figure 2. Independent constant-power burnup history versus the reported rounded exit target.

S5 printed initial k-difference values

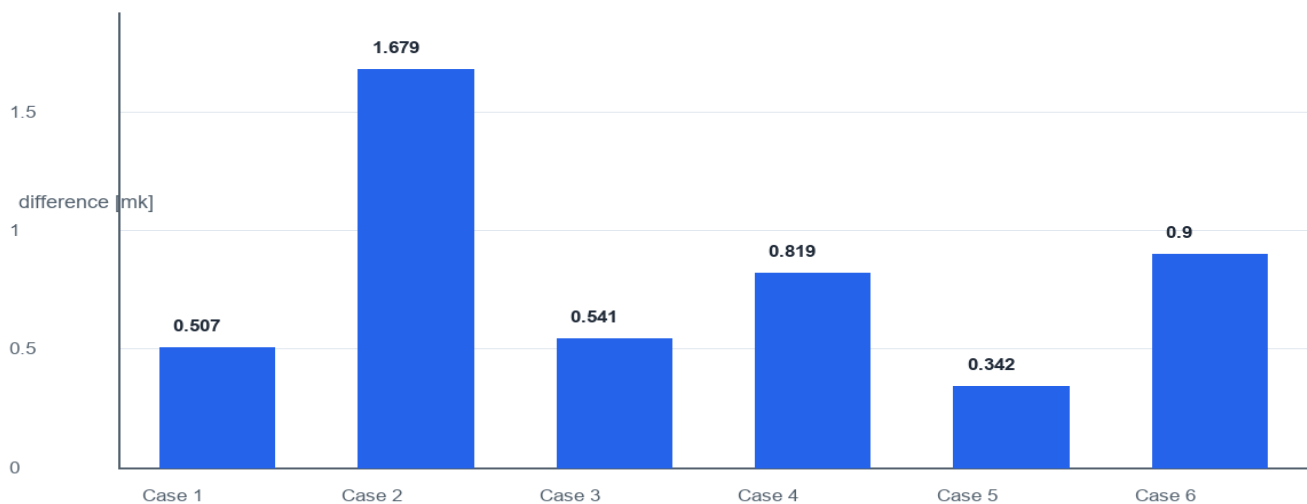


Figure 3. Six printed initial k-difference values retained as reported targets; the paper's below-2-mk statement is not converted into a tolerance.

Quantity	Value	Comparison
Power density	31.9713 kW/kg	Reported input
Duration	300 days	Reported input
Synthetic exit burnup	9.59139 GWd/t	ComputedTheory
Reported exit burnup	9.6 GWd/t	0.00861 GWd/t absolute difference (0.0897%)

4. Case C - S1 full-core schedule and alternate-cladding summary

The S1 source describes a 380-channel, 12-bundle/channel CANDU-6 model with alternating coolant directions, an 8-bundle shift, 4 channels/day, and a 300 full-power-day history. The synthetic history makes those schedule quantities complete and deterministic.

S1 synthetic full-core refuelling schedule

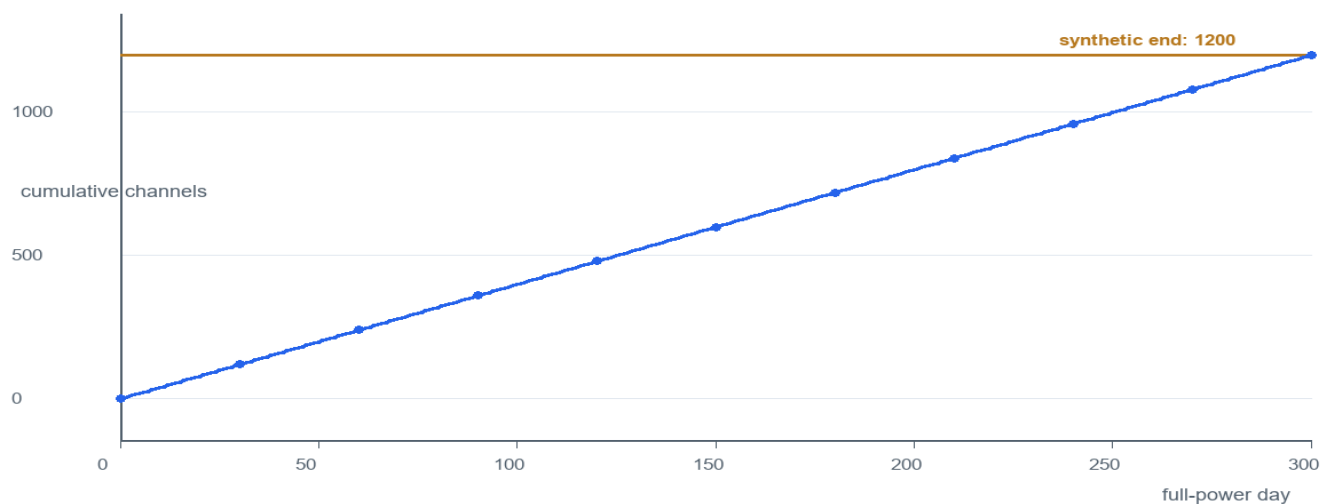


Figure 4. Complete synthetic cumulative refuelling schedule; the final 1200 channel events are 4 channels/day for 300 days.

S1 reported enriched-boron interval

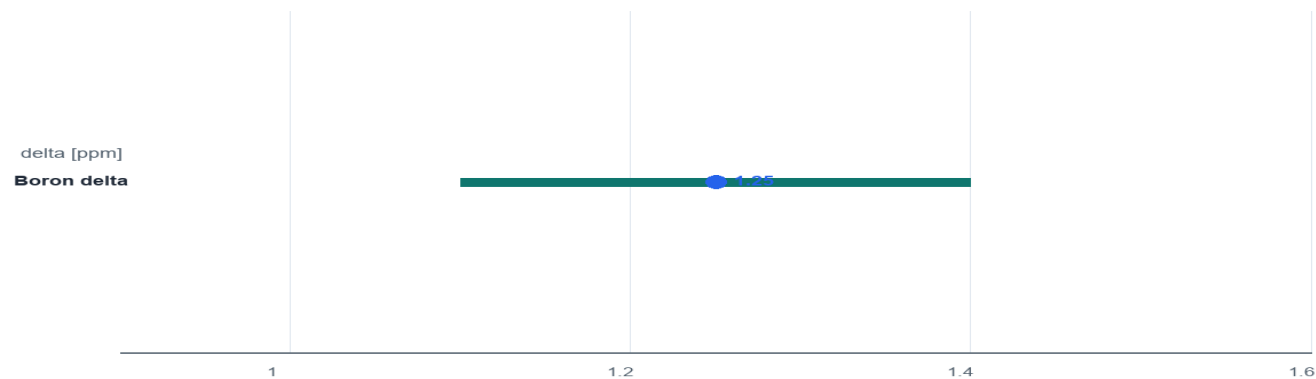


Figure 5. The synthetic +1.25 ppm boron value is placed at the midpoint of the reported +1.1 to +1.4 ppm range.

Observable	Reported target	Synthetic value	Disposition
SiC reactivity change	+3.4 mk	+3.4 mk	Reported-summary replay
Enriched boron change	+1.1 to +1.4 ppm	+1.25 ppm	Inside reported interval
Enriched channel power change	-246 kW	-246 kW	Reported-summary replay
History	380 channels; 12 bundles/channel; 8-bundle shift; 4 channels/day; 300 FPD	1200 channels; 9600 shifted bundles	Synthetic schedule

These values are deliberately not represented as extracted source curves or a runtime crosswalk. They are small reported targets paired with a deterministic project-authored schedule.

5. Case D - S4 full-core refuelling aggregate fixture

The Holmes thesis Table 5.2 compares a time-average reference with a reported simulation. The synthetic package replays the seven small simulation aggregates exactly and retains the time-average values so their differences remain visible.

S4 reported simulation vs time-average reference

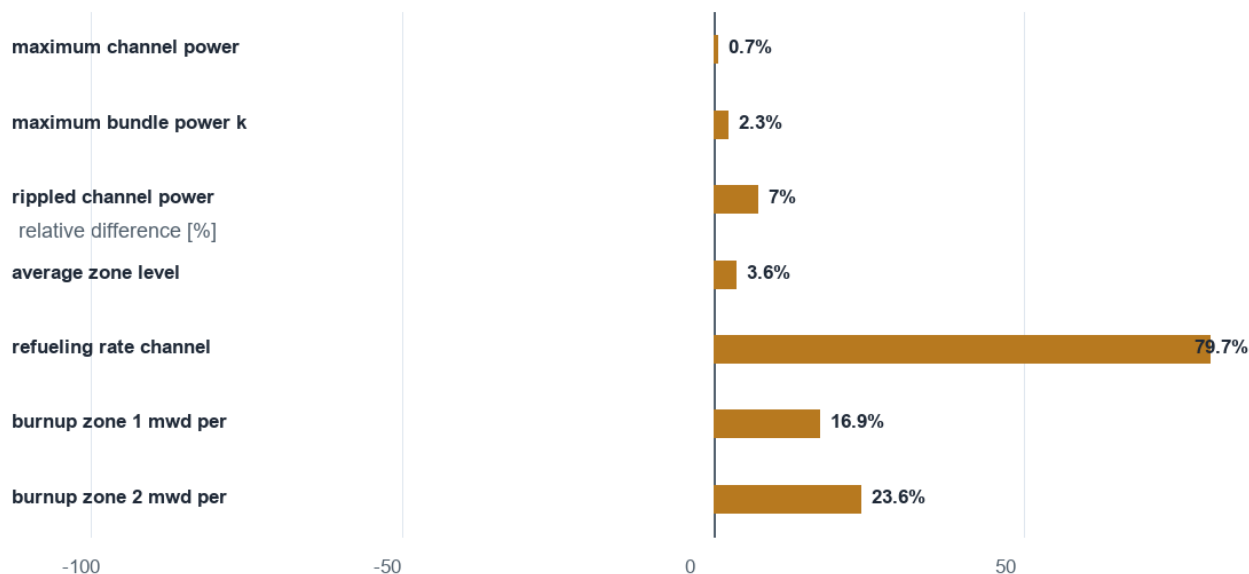


Figure 6. Relative difference between the reported simulation aggregate and the thesis time-average reference. The bars describe the source comparison; they are not solver error tolerances.

Metric	TA reference	Reported simulation	Synthetic	Sim-TA
maximum channel power kw	7,244	7,296	7,296	52
maximum bundle power kw	905	926	926	21
rippled channel power over reference	1	1.07	1.07	0.07
average zone level	0.5	0.482	0.482	-0.018
refueling rate channels per day	2.27	4.08	4.08	1.81
exit burnup zone 1 mwd per t	6,700	5,565	5,565	-1,135
exit burnup zone 2 mwd per t	5,360	4,095	4,095	-1,265

The deterministic schedule contains 408 refuelling events across 100 days, exactly 4.08 channels/day, with eight bundles per event. This is a reproducible provenance/table fixture; it is not a DONJON4 or CANFUEL run.

6. Reproducibility, validation, and next admission step

The standalone generator is independent of ReactorSim.Core. It reads the frozen definition, produces canonical UTF-8 JSON, validates exact artifact bytes and hashes, and checks the case invariants used by the focused Golden boundary tests. Candidate and approved artifacts differ only in their explicit disposition fields and are regenerated independently.

Check	Evidence
Definition binding	d93a8ddf8f34e184584f8a7718921503294bffacfa153b440191bd6beaefeb17
Approved artifact	b4d46cf2535c3616622c4c0a15ab1fcc6979662d3f1e02c25f4de0e1c61228ca / 211,470 bytes
Independent regeneration	True / b4d46cf2535c3616622c4c0a15ab1fcc6979662d3f1e02c25f4de0e1c61228ca
Cases and schedule	4 cases; 301-day S5 history; 301-day S1 schedule; 101-day S4 schedule; 7 S4 aggregate metrics
Physical tolerance	Not selected
Core mapping	Prohibited by artifact boundary

Required next step for any physical/external claim

A separate owner-authorized admission task must supply exact source build identity, nuclear-data files/checksums, complete geometry and state/history, units/normalization, output ordering, independent rerun evidence, and lawful artifact handling. Until then, the package is an authoritative synthetic regression artifact only.

References

- P1-T08 digest and source manifest in the repository.
- S1: <https://publications.polymtl.ca/5048/> (official PolyPublie record).
- S5: <https://publications.polymtl.ca/5047/> (official PolyPublie record).
- S4: https://publications.polymtl.ca/1307/1/2013_BradfordHolmes.pdf (official thesis PDF).